

Session 5.3

Pretrained Language Models: BERT, GPT, Alignment, and Five Limits

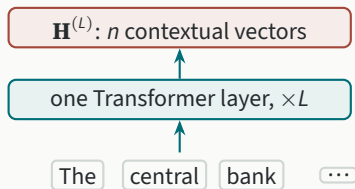
What pretraining buys, what alignment changes, and the five limits that follow from the architecture

Based on Chapter 13 of Miao & Feng, *AI for Economists*



Zhigang Feng

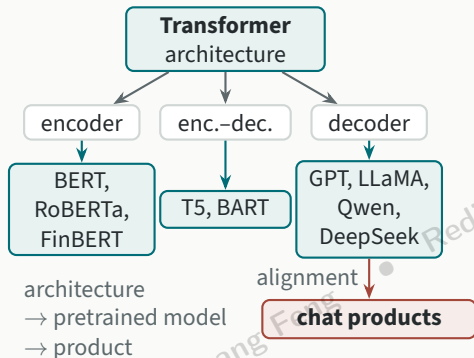
Where the Pipeline Stands



untrained, the attention row for *interest* runs 0.106–0.118: flat

- ▶ Deck 5.2 built one Transformer layer and stacked it L times: n tokens in, n contextual vectors $\mathbf{H}^{(L)}$ out.
- ▶ With random weights it reads nothing: the architecture holds no language yet.
- ▶ This deck: how the weights are fit on raw text; how the two model families measure; how a text generator becomes an assistant; which limits follow.
- ▶ **Pretraining puts language into the weights; how you use them decides the measure.**

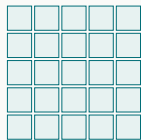
The Family Tree: Architecture, Pretrained Model, Product



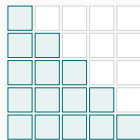
- **Architecture**: the wiring of Deck 5.2's layers, with random weights.
- **Pretrained model**: the same wiring with weights fit on a corpus. Encoders read; decoders write.
- **Product**: a pretrained decoder taught to follow instructions (**alignment**) and served with default settings.
- **Name the layer you mean: the product is not the architecture.**

One Block, Three Wirings

encoder: all
read all



decoder: i reads
 $1..i$



encoder-decoder: an encoder reads the input; a decoder writes the output, reading the encoder by cross-attention

- ▶ On “The central bank raised interest rates amid rising inflation”:
- ▶ **Encoder**: nine tokens in, nine contextual vectors out, each built from both sides.
- ▶ **Decoder**: “The central bank ... amid rising” in; a probability for every vocabulary token as the next one out.
- ▶ **Encoder-decoder**: the sentence in, another sequence out: a translation, or a record {action: raise, reason: inflation}.
- ▶ **The wiring fixes what the model can be asked for.**

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The Encoder Route

BERT: fill in blanks, pool, add a head, adapt to a domain



Pretraining an Encoder: Fill in the Blanks



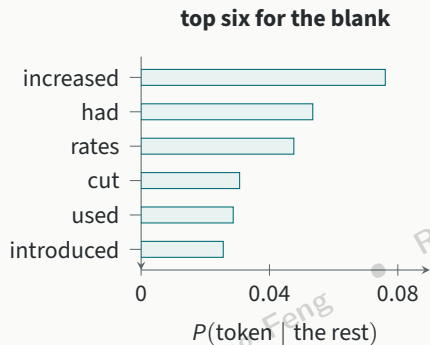
both sides needed: *raised? cut?*

choose 15% of positions; of those
80% $\rightarrow$ [MASK], 10% $\rightarrow$ a random
token, 10% unchanged

BERT also predicted whether sentence B follows A; RoBERTa found that dropping it “matches or slightly improves” results (Liu et al. 2019).

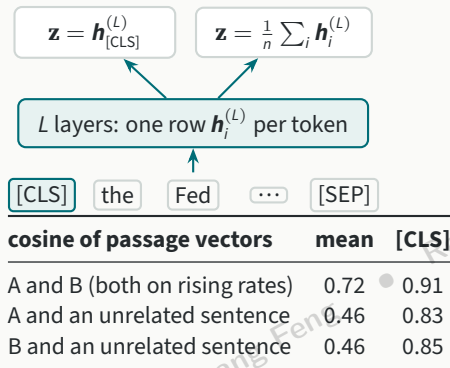
- ▶ Goal: fit the weights on raw text, no labels: hide words and let the text supply the answers.
- ▶ $\mathcal{M}$: the chosen 15% of positions; $\tilde{w}_{1:n}$: the corrupted input. [MASK] never appears in use, hence the 10% kept and 10% swapped (Devlin et al. 2019).
- ▶ Maximise $\sum_{i \in \mathcal{M}} \log P_{\theta}(w_i \mid \tilde{w}_{1:n})$, where $P_{\theta}(\cdot) = \text{softmax}(\mathbf{W}_{\text{out}} \mathbf{h}_i^{(L)})$ over the V tokens and θ is every weight.
- ▶ **To fill a blank the model must read both sides, so $\mathbf{H}^{(L)}$ encodes context.**

Example: A Pretrained Encoder Fills the Blank



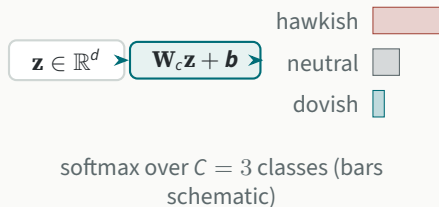
- ▶ Multilingual BERT's masked-word head, recorded offline, on “The central bank [MASK] interest rates amid rising inflation.”: *increased* 0.076, *cut* 0.031, the true *raised* 0.019, so this blank's loss is $-\log 0.019 = 3.97$.
- ▶ “Inflation (unemployment) is rising, so the Committee decided to [MASK] the target range”: $P(\text{raise})$ 0.0014 vs 0.0029, the wrong way round.
- ▶ **Pretraining delivers the kind of word, not the economics: hence domain text and labels.**

From Token Vectors to One Vector: [CLS] and Mean Pooling



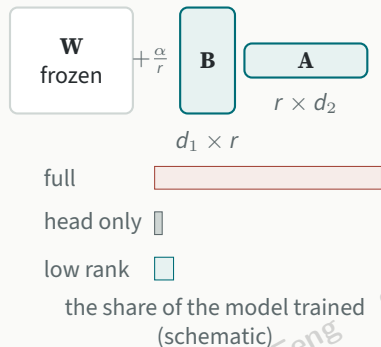
- ▶ Input: [CLS] $w_1 \dots w_n$ [SEP]. [CLS] is a slot with no word whose last-layer row stands for the text; [SEP] marks the end.
- ▶ Two ways to one vector, **pooling**: the [CLS] row, or the mean of all rows.
- ▶ Recorded on multilingual BERT: a policy statement (A) and an annual-report passage (B) score 0.72 by the mean, against 0.46 with unrelated text. Raw [CLS] rows are all alike.
- ▶ **Untuned, pool by the mean; the [CLS] row becomes useful once a task trains it.**

Fine-Tuning: A Classification Head on $\mathbf{z}$



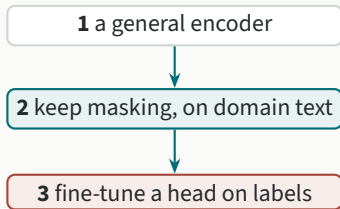
- ▶ Goal: a stance for each sentence, learned from N hand-labelled ones.
- ▶ Head: $\mathbf{p} = \text{softmax}(\mathbf{W}_c \mathbf{z} + \mathbf{b})$, $\mathbf{W}_c \in \mathbb{R}^{C \times d}$; loss $-\frac{1}{N} \sum_i \log p_{i, y_i}$, y_i sentence i 's label.
- ▶ A multinomial logit on the encoder's features; full fine-tuning also moves the encoder.
- ▶ Counts: the head has $C(d + 1) = 2,307$ numbers ($d = 768$); the encoder 177,853,440.
- ▶ **A few hundred labels can pin down the head, not the encoder.**

Three Ways to Fine-Tune



- **Full**: update all 177,853,440 weights. Closest fit; with few labels it overfits and loses the general representation.
- **Frozen encoder**: train the head alone. Cheap; the representation cannot adapt.
- **Low-rank adaptation** (LoRA; Hu et al. 2022): $\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r}\mathbf{B}\mathbf{A}$, $\mathbf{A}$ random, $\mathbf{B} = \mathbf{0}$ at the start, so training begins at the pretrained model. At $d_1 = d_2 = 768, r = 8$: 12,288 numbers against 589,824, 2.1%.
- **The change a task needs has low rank: one base model, one small versioned adapter per task.**

Domain-Adaptive Pretraining: Same Objective, Different Corpus

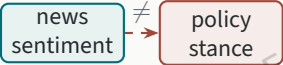


no new architecture at any step

- ▶ Continue the masking objective on text from the domain before adding a head (Gururangan et al. 2020).
- ▶ Why: *hawkish* in general text is a stance on war; in central-bank text it leans to raising rates.
- ▶ FinBERT (Araci 2019): BERT, then masking on TRC2-financial (46,143 Reuters articles, over 29M words), then a three-way head on the Financial PhraseBank (4,845 sentences).
- ▶ **Which corpus the model read is part of what the measure means.**

Example: A Domain Model's Labels Are Its Definition

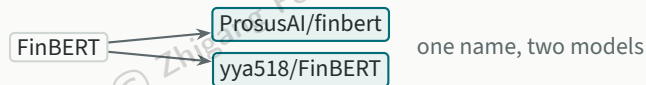
sentence, written for this slide	positive	negative	neutral
The Committee decided to raise the target range for the federal funds rate.	0.12	0.03	0.86
The Committee decided to lower the target range for the federal funds rate.	0.05	0.50	0.45
The Committee decided to maintain the target range for the federal funds rate.	0.06	0.03	0.91
The labor market has remained strong, and job gains have been solid.	0.95	0.03	0.02
Economic growth has slowed, and business investment has weakened.	0.01	0.97	0.02



- ProsusAI/finbert, recorded offline. A hike reads neutral (0.86), a cut negative (0.50): news for markets, not policy stance.
- **A classifier measures what its training labels meant: read the label set first.**

Two FinBERTs, and a Model With Two Classifiers

identifier	its paper	what it is
ProsusAI/finbert	Araci (2019)	financial news, then a three-way head on the Financial PhraseBank
yya518/FinBERT	Yang, UY & Huang (2020)	a different model of the same name, on financial communications
CentralBankRoBERTa	Pfeifer & Marohl (2023)	two classifiers: five macroeconomic agents, and a sign for each



- **Load a model by its identifier, cite the paper with it, and read its label set.**

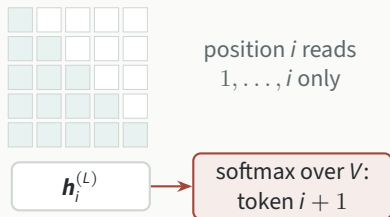
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The Decoder Route, and the Price of Scale

GPT: one next-token distribution, decoding, prompting, and what scale buys



Pretraining a Decoder: Predict the Next Token



- ▶ Same blocks with the causal mask: each row sees only its past, so it can predict what follows.
- ▶ $P_{\theta}(w_{i+1} \mid w_{1:i}) = \text{softmax}(\mathbf{W}_{\text{out}} \mathbf{h}_i^{(L)})$; **causal language modelling** maximises $\sum_i \log P_{\theta}(w_i \mid w_{<i})$.
- ▶ One nine-token sentence gives 8 predictions (The→central, The central→bank, ..., ... rising→inflation); masking 15% of it gives about 1.35.
- ▶ **A denser training signal, and only the left context: that trade separates GPT from BERT.**

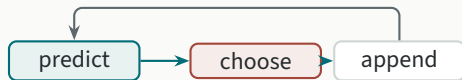
Example: A Next-Token Distribution, Recorded

next character	P	score
m	0.532	6.34
r	0.084	4.49
k	0.072	4.35
t	0.061	4.18
i	0.049	3.95
a	0.038	3.69

The Committee decided to_ m?

- ▶ **Machine B:** Session 1's text generator, trained on the 245 FOMC statements; 3 layers, $d = 128$, 631,424 parameters. Its tokens are 78 characters.
- ▶ After "The Committee decided to", it puts 0.53 on m , for *maintain*; then r (raise, reduce) and k (keep).
- ▶ Taking the top character each time: "... to maintain the target range for the federal funds rate at 0 to".
- ▶ **The model returns a distribution; a decoding rule turns it into text. Lab C4 runs it.**

Decoding: From a Distribution to Text



next	$\tau = 0.5$	$\tau = 1$	$\tau = 2$
m	0.93	0.53	0.19
r	0.02	0.08	0.07
k	0.02	0.07	0.07
t	0.01	0.06	0.06

- ▶ Generation loops: predict, choose one token, append it, predict again.
- ▶ **Greedy**: the top token. **Temperature**: $p_{\tau}(w) \propto \exp(s_w/\tau)$ on the scores s_w ; $\tau \rightarrow 0$ is greedy, $\tau > 1$ flattens. Top- k and top- p draw only from the best k tokens, or the smallest set with probability p .
- ▶ On Machine B's own scores (table): sampled at $\tau = 2$ it writes "neiqvelopments, both polifle advi2\$105. Biral 2 1.S5 fillp4".
- ▶ **For measurement set $\tau = 0$; report τ , the seed, the version and an input hash.**

In-Context Learning: The Task Goes in the Prompt

Label the monetary-policy stance of the Federal Open Market Committee statement below. Answer with exactly one word: HAWKISH if it leans toward tighter policy, DOVISH if it leans toward easier policy, NEUTRAL if it leans neither way. Output the single word and nothing else.

STATEMENT: [the statement of 1995-02-01: "The Federal Reserve Board today approved an increase in the discount rate from 4 3/4 percent to 5 1/4 percent, effective immediately." ...]

continuation

HAWKISH

recorded:
claude-sonnet-5,
2.1.278 (Claude
Code), 2026-09-20

- ▶ **Zero-shot**: an instruction and the item, as here. **Few-shot**: add worked examples first, e.g. "... decided to raise the target range. HAWKISH" (written for this slide).
- ▶ No parameter changes: the label is the model's continuation of the prompt.
- ▶ **The prompt is the estimator's specification: freeze it and publish it.**

In-Context Labels Carry No Internal Error Measure

estimation
fit to labels, then residuals $y_i - \hat{y}_i$



in context
an instruction and an item, one pass

nothing is fitted, so nothing inside reports an error

- ▶ Estimation fits parameters to labelled data and leaves regression residuals $y_i - \hat{y}_i$ to inspect.
- ▶ In-context labelling fits nothing: the model returns HAWKISH and no quantity says how likely it is to be wrong.
- ▶ Machine B's fluent statements had the same property: no target to subtract them from.
- ▶ **So the error is measured outside, on a hand-coded sample. Deck 5.4 builds it.**

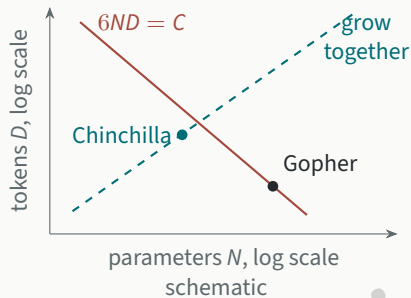
Two Paragraphs, Two Families

A, a policy statement: To counter rising inflation pressure, the central bank decided to raise the policy rate by 25 basis points. The Committee judges that maintaining a restrictive stance is appropriate until inflation returns to its target.

B, an annual report: During the reporting period, the company's financing costs increased as market interest rates rose. Management expects tighter credit conditions to put some pressure on next year's capital expenditure.

- ▶ **BERT**: pool each into $\mathbf{z}_A, \mathbf{z}_B$: cosine 0.72, against 0.46 with unrelated text; with labels, a stance head on $\mathbf{z}$.
- ▶ **GPT** (illustrative): "... the central bank decided to" continues *raise*; asked A's stance, "hawkish"; asked B's risk type, {risk type: rising financing costs}.
- ▶ *Rate* is the policy rate in A, a market rate in B; both read that from context.
- ▶ **BERT returns the same number every time; GPT moves with the prompt, decoding and version.**

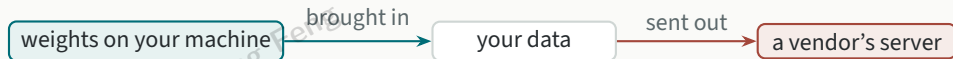
Scaling Laws, and What Chinchilla Corrected



- ▶ Held-out loss falls as a power law, $L(N) = (N_c/N)^{\alpha_N}$ with $\alpha_N \approx 0.076$: a straight line in logs over seven orders of magnitude (Kaplan et al. 2020).
- ▶ Hoffmann et al. (2022) fit $L(N, D) = E + A/N^{0.34} + B/D^{0.28}$ and minimise it subject to compute $C \approx 6ND$: double the parameters, double the tokens.
- ▶ Chinchilla, 70B parameters on 1.4T tokens (20 per parameter, $C = 5.9 \times 10^{23}$), beats Gopher, 280B on 300B ($1.1; 5.0 \times 10^{23}$).
- ▶ **A production function: parameters and tokens are inputs, $6ND$ is the isocost.**

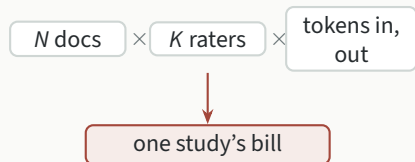
Open Weights, and Where the Data Is Allowed to Go

route	what you can archive	what it costs
a commercial interface	a version string, a call date and the sampling parameters	the text leaves the building, and the version moves underneath you
open weights, run locally	the weights themselves, beside the code and the prompt	your own hardware, and a step behind the frontier



- ▶ **Open weights:** the parameter files are published, so the model runs on your own hardware (LLaMA, Qwen, DeepSeek, GLM).
- ▶ **For confidential text the law decides, not accuracy: the model comes to the data.**

A Cost Formula, Not a Price List



per call it is nothing; per study it is a budget line

- ▶ $C = N K (t_{in} p_{in} + t_{out} p_{out})$: N documents, K raters or repeats, t tokens in and out per call, p the price per token on the day.
- ▶ In tokens: the lab's 245 statements average 425 tokens; with $K = 3$ and 5 tokens out, one pass reads and writes 316,050 tokens.
- ▶ Deck 5.5's replication of Yin, Vu & Persico (2026) runs 18,797 tasks with three raters.
- ▶ **Prices move every quarter: budget the study, and archive the day's quotation with the outputs.**

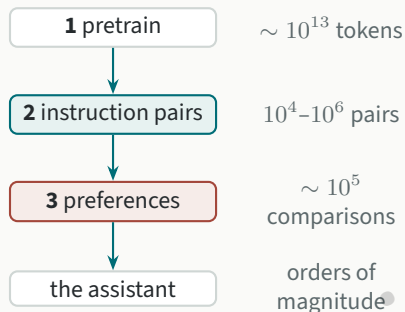
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Alignment

From a model that continues text to an assistant that answers



Why a Pretrained Decoder Is Not an Assistant



- ▶ Pretraining teaches continuation, not answers. Asked “How do I compute robust standard errors for a panel regression in Python?”, a base model may go on “Or: how do I do it in Stata...” (illustrative).
- ▶ Alignment adds two stages on far less data: fine-tuning on instruction–answer pairs, then learning from preferences.
- ▶ InstructGPT used about 13k prompts to fine-tune and 33k for its reward model (Ouyang et al. 2022).
- ▶ **Small stages that set how the model answers everything, your labels included.**

Stage 1: Supervised Fine-Tuning

x : Classify the stance of “The Committee decided to raise the target range.” One word.

y : Hawk ish .

each answer token scored given x and the earlier answer tokens (an example written for this slide)

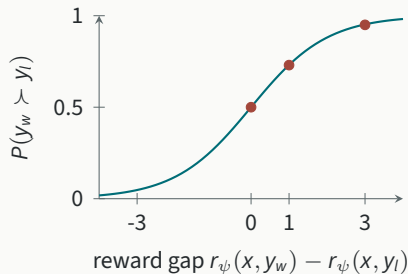
- ▶ Data: pairs (x, y) , an instruction x and a written answer $y = (y_1, \dots, y_m)$.
- ▶ Loss on the answer tokens only:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y)} \sum_{j=1}^m \log P_{\theta}(y_j \mid x, y_{<j}),$$

the next-token loss, on instruction data.

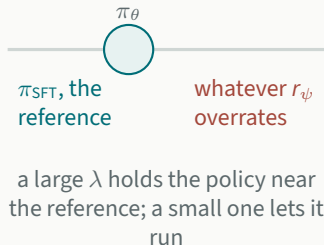
- ▶ It fixes form: format, when to decline, how far to reason.
- ▶ **Each instruction carries one answer, so the model never learns why another would be worse.**

Stage 2a: A Reward Model From Comparisons



- ▶ Annotators see two replies to one instruction x and pick one: y_w preferred to y_l .
- ▶ Bradley-Terry:
 $P(y_w \succ y_l | x) = \sigma(r_\psi(x, y_w) - r_\psi(x, y_l))$; r_ψ is the pretrained network with a one-number head, fit by minimising $-\sum \log \sigma(r_\psi(x, y_w) - r_\psi(x, y_l))$.
- ▶ Worked: reward gaps of 0, 1 and 3 give 0.50, 0.73 and 0.95.
- ▶ **A binary logit: utility r_ψ plus type-I extreme-value noise (Luce-McFadden). Deck 5.4 fits the same model to coded survey answers.**

Stage 2b: Optimise Against the Reward, on a Leash

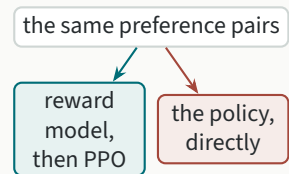


- ▶ The **policy** $\pi_\theta(y | x)$: the model's probability of reply y to prompt x .

$$\max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [r_\psi(x, y)] - \lambda \mathbb{E}_{y \sim \pi_\theta} \log \frac{\pi_\theta(y | x)}{\pi_{\text{SFT}}(y | x)}$$

- ▶ The second term is the KL divergence from the fine-tuned model; without it the policy finds replies r_ψ overrates: **reward hacking**.
- ▶ PPO (Deck 3.5) solves it, clipping the ratio $\pi_\theta / \pi_{\text{old}}$ to $[1 - \epsilon, 1 + \epsilon]$ so no step moves the policy far.
- ▶ **Ridge for a policy:** $\|y - X\beta\|^2 + \lambda \|\beta - \beta_0\|^2$ shrinks to β_0 ; here π_{SFT} is β_0 .

Direct Preference Optimisation: The Same Pairs, No Reward Model



two routes, one object: what the annotators preferred

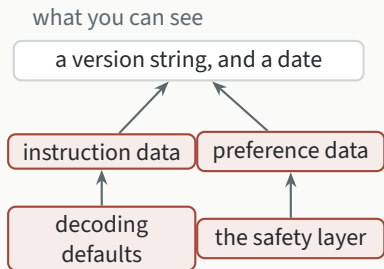
- ▶ The least problem's optimum:
 $\pi^*(y | x) = \pi_{\text{SFT}}(y | x) e^{r(x,y)/\lambda} / Z(x)$, $Z(x)$ the normaliser.
- ▶ Invert it, $r = \lambda \log(\pi^* / \pi_{\text{SFT}}) + \lambda \log Z(x)$, and put it in the Bradley-Terry gap, where $Z(x)$ cancels:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E} \log \sigma \left(\lambda \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{SFT}}(y_w|x)} - \lambda \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{SFT}}(y_l|x)} \right)$$

(Rafailov et al. 2023, who write β for λ).

- ▶ **Raise the preferred reply relative to the reference, lower the rejected one: the same object, no reward model. Llama 3 (Grattafiori et al. 2024) and Qwen2 (Yang et al. 2024) use it.**

The Version Is the Instrument



what moved underneath it: private,
rolling, and not announced

- ▶ Instruction and preference data are the vendor's, private and rolling: from outside you see a version string, such as `gpt-4-0613` OR `gpt-4o-2024-08-06`.
- ▶ Two studies that both “used GPT-4” may have used different instruments, and neither can tell.
- ▶ Deck 5.5 scores one frozen rubric with several raters and shows how far the readings move.
- ▶ **An aligned model used for scoring is a time-varying instrument: the version is part of the measure's definition.**

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Five Limits

What the architecture implies, and what no amount of scale removes

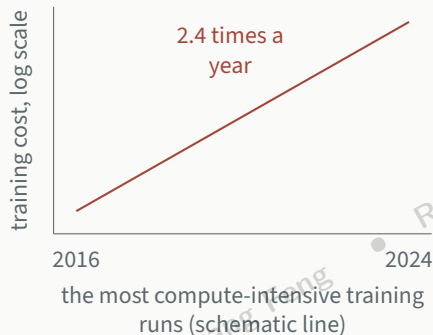


Five Limits, Five Architectural Roots

limit	root in the architecture	what it does to a design
the cutoff	knowledge sits in the weights	put the document in the prompt
hallucination	the objective rewards form	check every citation and number
the window	attention costs $O(n^2)$	blocks sever cross-references
lost in the middle	position bias in attention	position is part of the error
prompt sensitivity	the prompt is the prior	freeze it, and vary the order

- ▶ None is a defect awaiting a patch; each follows from how the machine is built.
- ▶ **A larger model moves all five, and removes none of them.**

Limit One: The Cutoff



- ▶ Everything the model knows sits in its weights, fixed on a date the vendor publishes. Later events are invisible to it.
- ▶ Updating knowledge means retraining, and the cost of the largest training runs has grown 2.4 times a year since 2016 (Cottier et al. 2025).
- ▶ **So the repair is not retraining: put the passage in the prompt. Session 6 makes that a system.**

Limit Two: Fluency Is Not Truth

what is maximised:

$$\log P_{\theta}(w_i \mid w_{<i})$$

a property of the *language*

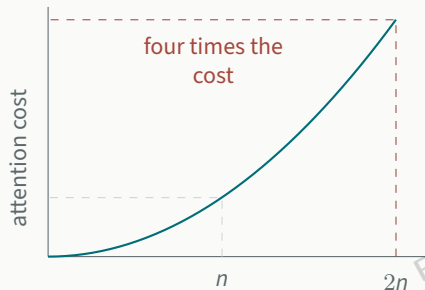
what you wanted:

a property of the *world*

where the training text was thin, the
form is filled with invention

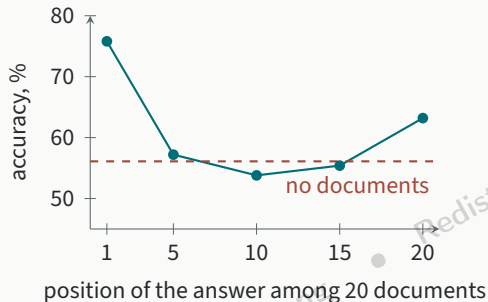
- ▶ **Hallucination:** fluent output that is false — invented citations, plausible numbers, laws that do not exist.
- ▶ The objective scores how likely the continuation is, never whether it is true.
- ▶ A citation's form, "Author (Year), *Title*, Journal 34(2): 101–120", is predictable token by token: every token can be likely while the whole is false.
- ▶ **Every citation, number and quotation is a draft until it is checked outside the model.**

Limit Three: The Window Is Quadratic



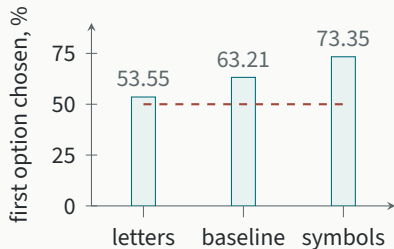
- ▶ Attention scores every position against every other: cost $O(n^2)$. A median FOMC statement of 410 tokens already needs 168,100 scores per head and layer.
- ▶ The window is a hard bound: one annual report fits, a decade of transcripts does not.
- ▶ Cutting a document into blocks severs the reference from the management discussion back to the risk factors.
- ▶ **Sparse attention and state-space models (Gu & Dao 2023) cut the cost and give up exact global attention.**

Limit Four: Lost in the Middle



- ▶ Move the document that holds the answer through 20 retrieved documents: accuracy traces a U, best at the ends, worst in the middle (Liu et al. 2024; GPT-3.5-Turbo, their Table 6).
- ▶ In the middle it does worse than with no documents at all: more context is not more information.
- ▶ Base models show the same curve.
- ▶ **Non-classical error: in long and short filings the noise differs, by position.**

Limit Five: The Prompt Is the Prior



- ▶ Where order should not matter, GPT-4 picked the first option 63.21 per cent of the time; 53.55 with letter labels, 73.35 with symbols (Brucks & Toubia 2025).
- ▶ Saying the order was randomised does not repair it: 63.82 against 65.00 per cent.
- ▶ The same pull towards the ends as the long-context study, on a list of labels.
- ▶ **Freeze the prompt, version it, publish it, vary the order — Deck 5.4 turns all five limits into a protocol.**